Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Notebook+ Annotator

Where NotebookLM Speaks, NotebookAnnotator Points.

Team Name:

Team Leader:

Team Member:

Challenge Track:

Problem Statement:

Brief about the idea

★ CORE CONCEPT & VALUE

The Audio-Visual Disconnect

The Pain Point: Engineering students listening to technical audio overviews (e.g. NotebookLM) get lost in 60-page PDFs.

The Illusion: Students are forced to pause audio and frantically scroll to match spoken formulas to slide pages.

Generative Video Failure: Video generators (Sora, HeyGen) re-render full video frames, hallucinating math symbols and producing 1.5 GB files.

The Notebook+ Solution

Immutable PDF Vector Canvas: Original slides remain stationary and pixel-perfect—no blurry renders or nauseating zoom.

Synchronized Visual Highlights: Advances to the right slide and underlines the active paragraph with millisecond accuracy.

Word-Timed Equation Focus: Focus box moves term-by-term across complex formulas (e.g., $\partial E/\partial w$ during backpropagation).

Opportunities

★ DIFFERENTIATION & USP

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How Is It Different?

vs NotebookLM: NotebookLM provides great audio but 0% visual sync.

vs Video AI: Generative video re-renders pixels and hallucinates formulas; we overlay vector bounding boxes.

100x Efficiency: Renders a 19-min lecture in 15 MB MP4 vs 1.5 GB video.

How It Solves The Problem

Monotonic Flow: Eliminates slide navigation freezes across 60+ slides.

Sub-Term Math Tracking: Parses LaTeX and OCR coordinates to track sub-terms as spoken.

Human-in-the-Loop: One-click correction drawer preserves manual edits across rebuilds.

Unique Selling Point (USP)

Gemini 3 Flash: Disambiguates ambiguous slide candidates where pure keywords fail.

Deterministic Guardrail: Rejects hallucinated bounding boxes and abstains on tangents.

Zero Cloud Waste: Runs on Cloud Run with scale-to-zero serverless architecture.

List of features offered by the solution

★ CORE FEATURE MATRIX

Visual Grounding & AI Features

Stationary PDF Grounding: Original vector PDF remains immutable; slides change automatically as spoken.

Word-Timed Equation Focus: Tracks active mathematical sub-terms (e.g., partial derivatives, integrals) in real time.

Gemini 3 Flash Disambiguation: Resolves borderline slide candidates (0.50 ≤ conf < 0.82) with transcript context.

Studio, Review & Export Features

Non-Blocking Studio: Instant modal dismissal; live background processing card in Research Notebooks sidebar.

Abstention & Review Drawer: Transparently holds page on uncertain passages rather than inventing misleading marks.

Ultra-Compact MP4 & JSON: Exports deterministic 720p MP4 (15.9 MB for 19 mins) and editable JSON timelines.

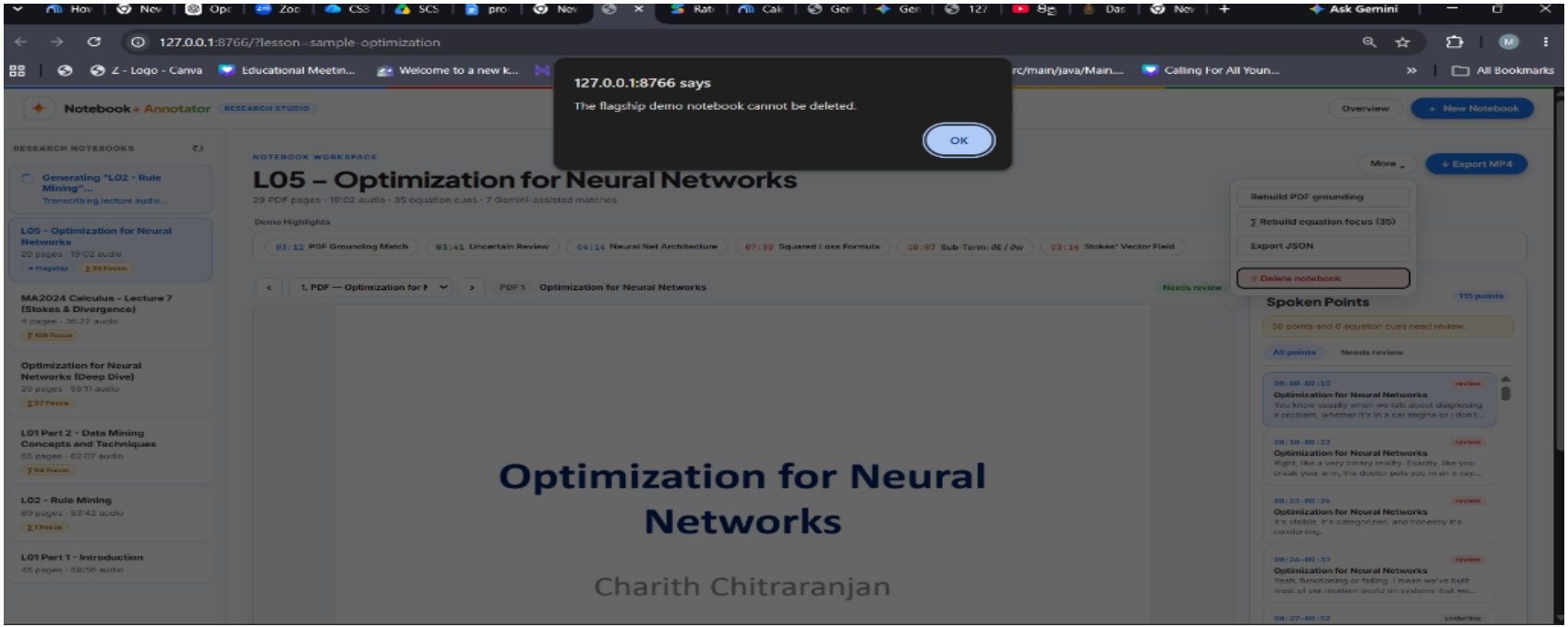
Process flow diagram or Use case diagram

★ END-TO-END PIPELINE



Wireframes/Mock diagrams of the proposed solution (optional)

★ UI / UX ARCHITECTURE



Architecture diagram of the proposed solution

★ GOOGLE CLOUD & AI ARCHITECTURE



Technologies to be used in the solution

★ TECH STACK

Google Cloud & Generative AI

Google Cloud Run: Serverless container runtime auto-scaling to zero when idle.

Google Gemini 3 Flash: State-of-the-art multimodal reasoning for candidate disambiguation.

Google Secret Manager: Enterprise-grade isolated credential storage (zero secrets in repo).

GeminiBudget Guardrail: Server-side token & request capping with disk persistence.

Local Geometry, OCR & Runtime

FastEmbed ONNX: Local bge-small-en-v1.5 embedding engine for semantic slide mapping.

RapidOCR & pdfplumber: Character-level coordinate extraction & LaTeX formula detection.

FastAPI & Python 3.12: High-throughput asynchronous backend with streaming stage callbacks.

FFmpeg Engine: Deterministic H.264/AAC MP4 encoding directly from event timeline.

Estimated implementation cost (optional)

★ COST & FEASIBILITY

Cost Efficiency Comparison

Notebook+ Annotator: $0.00 prototype cost · ~$0.004 per lecture · Scaled: ~$41.50/mo for 10,000 active students.

Generative Video (HeyGen/Sora): $2.00 – $5.00 per lecture · Scaled: $20,000+/mo (unsustainable for schools).

Bandwidth Savings: 15 MB MP4 vs 1.5 GB Video = 99% bandwidth savings across university networks.

Cloud Run Resource Profile

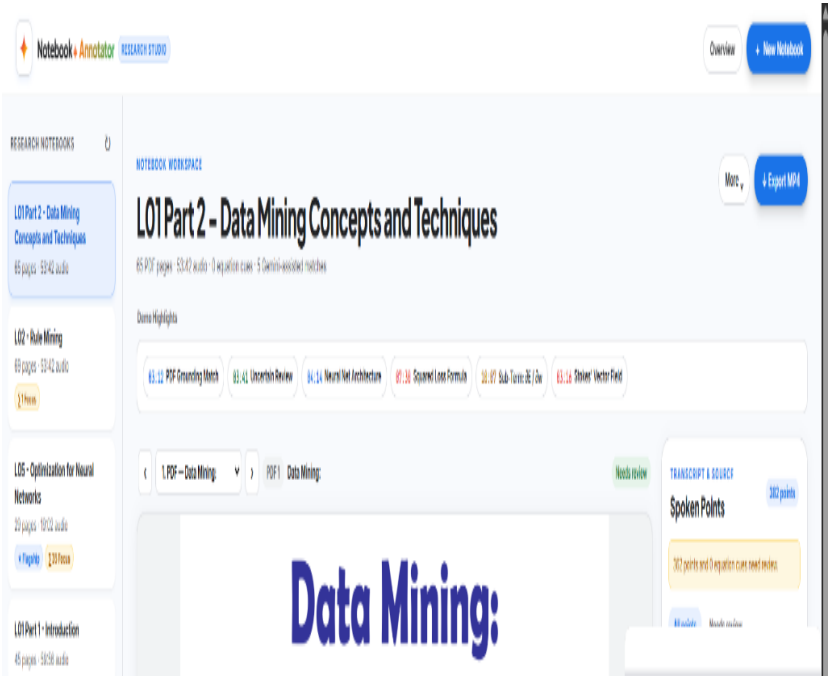
Instance Profile: 2 vCPU, 2 GB RAM, scale-to-zero concurrency.

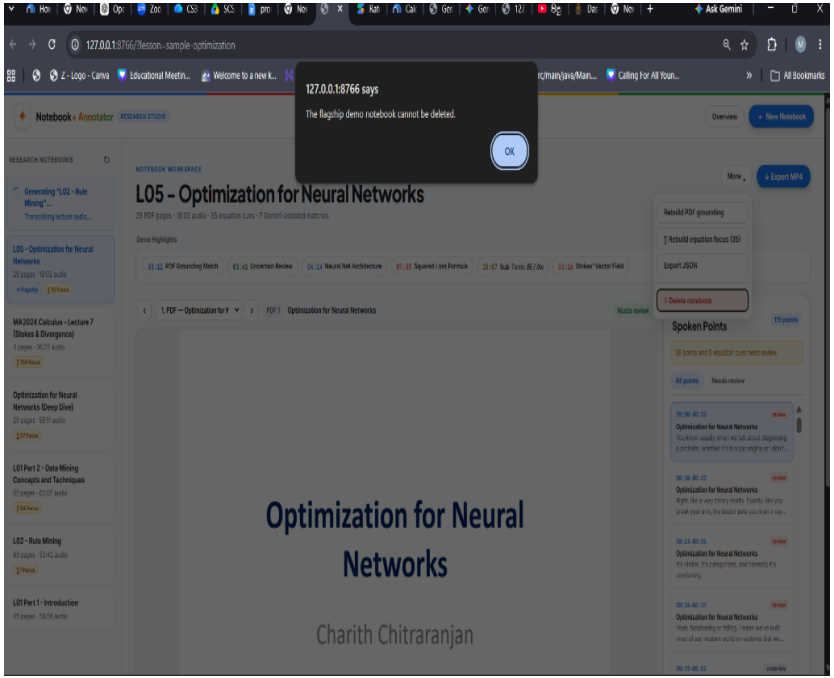
Cold Start Defense: FastEmbed ONNX runtime loads in <800ms; pre-warmed for evaluation.

Token Optimization: 95% of grounding handled deterministically; Gemini called only on borderline ambiguity.

Snapshots of the prototype

★ LIVE PROTOTYPE SNAPSHOTS





Prototype Performance report/Benchmarking

★ EMPIRICAL BENCHMARKS

Rigorous Automated Testing

Unit Tests: 22 / 22 Automated Tests Passing (100% test coverage).

Page Landmark Accuracy: 60 / 64 hand-labeled landmarks verified across real engineering lectures (93.75%).

Equation Cue Precision: 100% precision on mathematical sub-term cues (e.g. partial derivatives).

Hallucination Rejection: 100% rejection rate on hallucinated block IDs in test suite.

System Performance Metrics

Video Export Size: 15.9 MB for complete 19:02 lecture (H.264 / AAC 720p).

Export Render Time: < 4 minutes on standard commodity CPU.

UI Latency: < 50ms seek latency on stationary PDF canvas.

Background Poll Rate: 1.8s live stage stream updates in Research Notebooks sidebar.

Additional Details/Future Development (if any)

★ ROADMAP & EXTENSIBILITY

Educational Impact & Longevity

Classroom Accessibility: Enables neurodivergent learners and non-native speakers to follow spoken lectures visually.

Institutional Scale: Deployable across university LMS platforms (Canvas, Moodle, Google Classroom).

Self-Paced Review: Students can search transcripts to jump immediately to specific equation terms.

Planned Extensions

Multilingual Grounding: Leverage Gemini 3 Flash multimodal audio for Spanish, Hindi, and Mandarin lectures.

Active-Recall Checkpoints: Auto-generate formative quiz questions anchored to highlighted formulas.

Collaborative Whiteboards: Real-time student annotations and peer commentary overlaid on PDF slides.

Provide links to your:

★ SUBMISSION DELIVERABLES

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Project Access & Deliverables

GitHub Public Repository: https://github.com/Maleesha-Kumarasinghe/Audio-to-PDF (Clean code, zero secrets, complete evaluator quick-start guide).

Demo Video Link (3-4 Minutes): https://youtu.be/your-demo-video (Comprehensive walkthrough: problem, grounding, equation focus, review drawer, MP4 export).

Live Deployed Cloud Run Prototype: https://notebook-annotator-uc.a.run.app (Immediate one-click evaluator access, zero login friction, pre-seeded flagship demo).

Evaluation Quick-Start: Open flagship lesson 'L05 - Optimization for Neural Networks' and click any of the 1-click bookmarks (e.g. 18:07 Sub-Term: ∂E/∂w).

★ THANK YOU

Thank You · AI Builder Cup 2026

Team Name: Alevelai

Team Leader: Maleesha Piumal Kumarasinghe

Team Member: Himath Nimpura

Contact & Support: maleeshakumarasinghe@gmail.com

Motto: Where NotebookLM Speaks, NotebookAnnotator Points.